a

Cells

1 **Reporter cell lines**
tag brightness tracks expression

47 cell lines
SU-DHL-4 B cell lymphoma

## 1 47 reporter genes, by pathway

Each gene has its own tagged cell line and screen. Groups follow Fig. 2A.

**BCR signalling**
B cell receptor, 15 genes

| | | |
|---|---|---|
| BLK | CD45 | IGH |
| BLNK | CD79A | IGK |
| BTK | CD79B | LYN |
| CD20 | CD81 | PLCG2 |
| CD27 | CSK | SYK |

**PI3K-AKT-mTOR**
pathway, 6 genes

| | |
|---|---|
| AKT2 | MCL1 |
| BCL-XL | PIK3AP1 |
| BCL2 | PTEN |

**NF-κB signalling**
2 genes

| | |
|---|---|
| MAP3K7 | TNIP1 |

**Nuclear proteins**
and transcription factors, 24 genes

| | | | |
|---|---|---|---|
| BACH2 | ID3 | MYC | RB1 |
| BCL11A | IKZF1 | NCOA3 | REL |
| BCL6 | IRF8 | OCAB | SPI1 |
| BTG2 | MEF2B | OCT2 | SPIB |
| E2F2 | MSH2 | PARP1 | TCF3 |
| HDAC1 | MSH6 | PAX5 | TP53 |

b

Screen

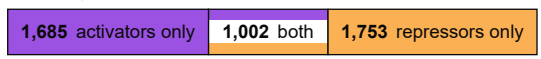
2 **CRISPR screens**
knock out genes, one per cell; sort the dimmest and brightest 5%

4,440 regulators
17,638 links

## 2 Regulators found by the screens

A regulator is a gene whose loss changes the brightness of a reporter.

**4,440 regulators**

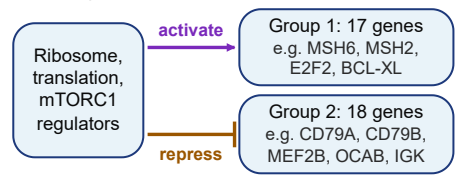



**17,638 links** a regulator and a reporter gene it changes




Both: activator for some reporters, repressor for others.
The box at right is a subset of these.

**Same regulators, opposite effects**




Groups 1 and 2: text and Fig. 5A.

c

Analysis

3 **Block mTORC1, GSK3 or both**
for 2 days, then RNA-seq

2,458 rapamycin
427 LY2090314
genes changed

**Key**

- step or pathway
- gene or gene group
- count

**b: regulator role**

- activator
- repressor
- both roles

**c: RNA after a drug**

- up
- down
- n.c. not listed as changed

## 3 mTORC1 and GSK3 often move the same genes in opposite directions

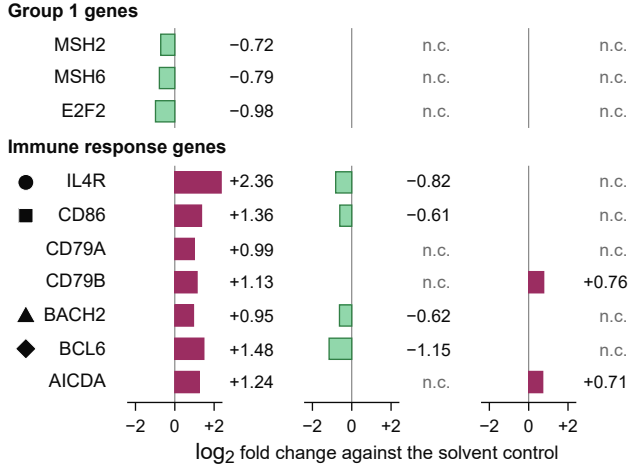
RNA-seq: change in a gene's RNA after the drug. +1 is twice as high, −1 half.
Reporter brightness (a, b) and RNA (c) are different measurements.

Genes the paper names as changed by rapamycin

**mTORC1 blocked**
rapamycin

**GSK3 blocked**
LY2090314

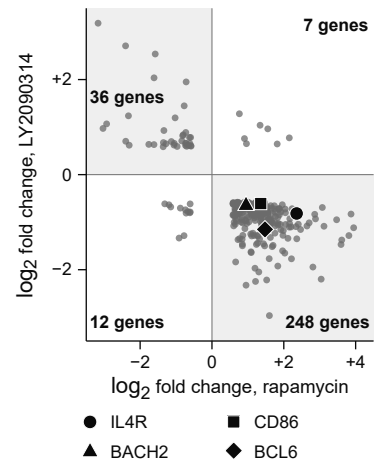
**Both blocked**
both drugs




With both drugs, 8 of these 10 genes are n.c.
Of 99 translation genes raised by rapamycin, 74 stay up with both.

**Genes changed by each drug**

284 of 303 genes in both lists move in opposite directions.




Shaded: opposite directions. Changed, by the paper's rule: base mean above 100, fold change above 1.5, adjusted p below 0.01.

Counts computed from the paper's tables; arrows follow its text.

After Figs 2A, 5, 7 of Kalchschmidt et al., Cell Rep. 2025, doi:10.1016/j.celrep.2025.116361, CC BY 4.0. Data: Tables S1, 3, 5, 6, 9, 10. Redesigned; values as published.